

Osman Akar

oak@math.ucla.edu | 424-535-7426

765 Weyburn Pl, Apt 325, Los Angeles, CA, 90024

LinkedIn://osmanakar | Website: www.math.ucla.edu/~oak

EDUCATION

PhD in Applied Mathematics

2019 - Current

University of California, Los Angeles

Research Interests:

- Computational Mathematics
- Numerical PDEs
- Physics Based Simulation for Graphics

Master of Arts in Mathematics

2016 - 2019

University of California, Los Angeles

- Departmental Scholars Program

Bachelor of Science in Mathematics of Computation

2015 - 2019

University of California, Los Angeles

- Math Undergraduate Merit Scholar
- Graduated Cum Laude

LANGUAGES

- Turkish: Native
- Russian: Beginner

PROGRAMMING

Python • C++ • C • MATLAB • $\LaTeX$

Familiar:

JavaScript • Assembly • Bash • Lisp

ACTIVITIES

- Los Angeles Math Circle

RESEARCH & WORK EXPERIENCE

UCLA IPAM | Research Intern at RIPS GumGum Team

JUNE 2018 – AUG 2018 | LOS ANGELES, CA

- Worked on a research team that build automated video augmentation pipeline that identifies crowd regions in sport videos and overlays an advertisement onto these regions, so that the advertisement looks aesthetic and natural for human perspective. This video link is an example of the final product. More explicitly, the automated video augmentation pipeline is able to 1) identify non-homogeneous textures (crowd regions in stadium imagery), 2) overlay a rectangular image asset on said region, and 3) constrain the asset to be placed in a perspective correct way for an overall enhanced user experience.
- Designed and implemented an algorithm to visualize RGB-Depth image in 3D.
- Developed a novel metric on video frames for the evaluation of the Pyramid Scene Parsing Network segmentation algorithm.
- Researched and experimented with various algorithms for focal length approximation. Optimized the focal length algorithm, reducing error percentage from % 38 to %22.
- The research paper "Augmented Reality for Non-homogeneous Textures" won "The Best Poster Award" in 1st International Workshop on Advanced Machine Vision for Real-life and Industrially Relevant Applications (AMV2018).

STAR LEAGUE, INC | Lecturer & Curriculum Designer (Part Time)

SEPTEMBER 2017 – CURRENT | IRVINE & SAN JOSE, CA

- Lectured math olympiad courses in AMC 8/10/12 and AIME levels throughout 12-weeks semester long sessions and 6-weeks summer camps. Presented solutions to problems from earlier years' math olympiads competitions.
- Organized curriculum and problem set for the 3-weeks Advanced AIME summer program. The problem set consists of about 650 problems with solutions and lecture notes.

UCLA APPLIED MATHEMATICS REU | Undergraduate Researcher

JUNE 2017 – AUGUST 2017 | LOS ANGELES, CA

- Worked on a research team that applied image processing techniques to extract non-content-based features from LAPD body-worn-camera video recordings and used machine learning to classify the recordings into categories.
- Created color-based feature and its metric system for video frames using CIELAB color space and Earth Mover Distance metric.
- Modified the old machine learning model to accept the new dataset of LAPD videos, experimented with tuning hyperparameters, and tested algorithm for different parameters. Reduced the error from %10 to %5.
- The research paper is submitted to arXiv:1904.09062.

HONORS & AWARDS

- Putnam Performance Prize by UCLA Mathematics Department, 2019
- Honorable Mention at William Lowell Putnam Mathematical Computation, 2018.
- Richard F. Arens Putnam Scholars Undergraduate Award, 2016
- Sole recipient of UCLA Math Undergraduate Merit Scholarship in class of '19.
- Gold Medal at International Mathematical Olympiad, 2014, South Africa
- Special Prize at European Union Contest for Young Scientist, 2014, Poland
- Silver Medal at International Mathematical Olympiad, 2013, Colombia